BRIAN M SCHILDER, PHD

Passionately pursuing transdisciplinary research to advance human health and knowledge.

STANDARD MODEL BIO

Genomic AI Scientist

Below are selected subsets of the full CV. -

EDUCATION

2024

Imperial College London / The Alan Turing Institute
PhD; Computational Genomics & Machine Learning — London, UK

Thesis: Multi-omic medicine: dissecting the cell type-specific and pleiotropic mechanisms underlying disease genomics at scale

2017

The George Washington University / Georgetown University
MPhil; Comparative Neuroscience & Genomics — Washington, DC, USA

Thesis: The evolution of the hippocampus and adult neurogenesis: novel insights into the origins of human memory

2011

Brown University / Princeton University
ScB; Neurological Diseases & Disorders — Providence, RI, USA

CORE SKILLS

Research

17+ years of deep expertise in genomics, AI, evolutionary biology and biomedicine. Strategically fuses concepts and methods across multiple domains.

- **Publication record**: 28 publications, 7 preprints and 14 awarded grants.
- **Reproducibility**: Global leader in promoting and enabling reproducible scientific practices. Writes 100% reproducible manuscripts programmatically.
- **Bioinformatics**: Created 46 Python and R packages to address key challenges in biological research.
- **High-performance computing**: Highly parallelised analyses and AI model training (CPUs and GPUs).
- **Web development**: 6+ websites, web apps, and interactive reports.

CONTACT

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UK
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LinkedIn
ORCID
Google Scholar
GitHub
Twitter
YouTube
Personal Website
Lab Website

SUMMARY

- 17+ years of research
- 28 publications
- 7 preprints
- 42 software packages
- 11 databases & apps
- 24 talks
- 15+ years of teaching & team management

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AI & Machine Learning

Proficient in developing and deploying AI/ML models (PyTorch, Keras, sklearn) to solve complex biological problems. Applied examples include:

- **Personalized VEP**: Adapted protein (ESM, AlphaFold), RNA (SpliceAI), and DNA (Flashzoi, phyloGPN, HyenaDNA) sequence models to generate personalized variant effect predictions for >100k clinical variants across >3,800 diverse human genomes. Demonstrated that sequence-based AI models capture biologically meaningful inter-individual variation, paving the way for personalized medicine.
- **TimelineTransformer**: Designed and trained an LLM-based time-series foundation model to predict patient health trajectories. Trained on 6.22M patient events.
- **Phenotype Severity Scoring**: Developed GPT-4-based framework to systematically score severity of 17k+ disease phenotypes to help reprioritization.
- **GenForge**: Developed transformer-based encoder-decoder hybrid model to reveal joint latent representation of genomic signatures across all diseases, phenotypes, pathways, and drugs.
- **MedAgent**: A reproducible multi-agent system for summarizing and interpreting multimodal medical data, including EHR, imaging, and genomics.
- **ReporterSuite**: Created proprietary Python packages for advanced LLM-based topic modelling of the PubMed literature to provide business intelligence to the world's largest digital health, biotech, and pharma companies (as a consultant with 120/80 Group).
- **echolocatoR**: Used functional impact predictions from DNA sequence models (DeepSEA, Basenji, IMPACT) to validate SNPs prioritised with Bayesian fine-mapping.

Project Management

Efficient management strategies to define objectives, track progress and coordinate diverse teams.

- **Documentation**: Defines objectives and tracks progress with GitHub Projects. Includes useful documentation in Issues, inline code and shareable reports.
- **Version control**: Extensive and daily use of GitHub, containers (*Docker, Singularity, virtual machines*), environments (*conda*) and pipelines (*Nextflow*).
- **Team management**: Led numerous collaborative research projects and supervised researchers at various career stages.

Soft Skills

Advances science through effective problem formulation, collaboration and communication.

- **Problem formulation**: Rapid hypothesis generation, project design, and creative problem solving.
- **Collaboration**: Diverse and global collaborative networking.
- **Communication**: Clear and concise distillation of complex results to a variety of audiences. Presented **25** conference posters.

PUBLICATIONS

2026 **Gene expression patterns of the developing human face at single cell resolution reveal cell type contributions to normal facial variation and disease risk**

Nature Communications (2026) https://doi.org/10.1038/s41467-026-70232-6

N Khouri-Farah, EW Winchester, **BM Schilder**, K Robinson, SW Curtis, NM Skene, E Leslie-Clarkson, J Cotney

2025 **Mondo: integrating disease terminology across communities**

Genetics (2025) https://doi.org/10.1093/genetics/iyaf215

NA Vasilevsky, … **BM Schilder**, …,PN Robinson, CJ Mungall, A Hamosh, MA Haendel

2025 **Fine-mapping genomic loci refines bipolar disorder risk genes**

Nature Neuroscience (2025) https://doi.org/10.1038/s41593-025-01998-z

M Koromina, A Ravi, G Panagiotaropoulou, **BM Schilder**, … S Ripke, T Raj, JRI Coleman, N Mullins

2025 **CUT&Tag recovers up to half of ENCODE ChIP-seq peaks**

Nature Communications (2025) (16):2993; https://doi.org/10.1038/s41467-025-58137-2

L Abbasova, P Urbanaviciute, D Hu, JN Ismail, **BM Schilder**, A Nott, NG Skene, SJ Marzi

2025 **Chromatin Interaction and Histone Mark Signatures Associated With TBXT Expression in Metastatic Lung Cancer**

Genes Chromosomes Cancer, (2025) (64):e70041; https://doi.org/10.1002/gcc.70041

RM Yaa, **BM Schilder**, RD Acemel, FC Wardle

2023 **rworkflows: automating reproducible practices for the R community**
Nature Communications (2023) 15(149); https://doi.org/10.1038/s41467-023-44484-5
BM Schilder, AE Murphy, NG Skene
📰 **News**
- Featured in *Nature Communications* Editors' Highlights

2023 **Artificial intelligence for neurodegenerative experimental models**
Alzheimer's & Dementia (2023) http://doi.org/10.1002/alz.13479
SJ Marzi, **BM Schilder**, A Nott, C Sala Frigerio, S Willaime-Morawek, M Bucholc, DP Hanger,C James, PA Lewis, I Lourida, W Noble, F Rodriguez-Algarra, JA Sharif, M Tsalenchuk, LM Winchester, U Yaman, Z Yao, DEMON Network, JM Ranson, DJ Llewellyn

2023 **Artificial intelligence for dementia genetics and omics**
Alzheimer's & Dementia (2023) http://doi.org/10.1002/alz.13427
C Bettencourt, NG Skene, S Bandres-Ciga, E Anderson, LM Winchester, IF Foote, J Schwartzentruber, JA Botia, M Nalls, A Singleton, **BM Schilder**, J Humphrey, SJ Marzi, CE Toomey, A Al Kleifat, EL Harshfield, V Garfield, C Sandor, S Keat, S Tamburin, C Sala Frigerio, I Lourida, DEMON Network, JM Ranson, DJ Llewellyn

2023 **Artificial intelligence for dementia research methods optimization**
Alzheimer's & Dementia (2023) http://doi.org/10.1002/alz.13441
M Bucholc, C James, A Al Khleifat, A Badhwar, N Clarke, A Dehsarvi, CR Madan, SJ Marzi, C Shand, **BM Schilder**, S Tamburin, HM Tantiangco, I Lourida, DJ Llewellyn, JM Ranson

2023 **EpiCompare: R package for the comparison and quality control of epigenomic peak files**
Bioinformatics Advances (2023) 13(1):vbad049; https://doi.org/10.1093/bioadv/vbad049
S Choi, **BM Schilder**, L Abbasova, AE Murphy, NG Skene

2022 **Dissecting the Shared Genetic Architecture of Suicide Attempt, Psychiatric Disorders, and Known Risk Factors**
Biological Psychiatry (2022) 91(3):313-327; https://doi.org/10.1016/j.biopsych.2021.05.029
N Mullins, J Kang, AI Campos,…**BM Schilder**, et al.

2022 **Genetic analysis of the human microglial transcriptome across brain regions, aging and disease pathologies**
Nature Genetics (2022) https://doi.org/10.1038/s41588-021-00976-y
K de Paiva Lopes, G JL Snijders, J Humphrey, A Allan, M Sneeboer, E Navarro, **BM Schilder**…T Raj
📰 **News**
- Microglial transcriptomics meets genetics: new disease leads (Nature Reviews Neurology, 2022)
- Mighty MiGA: Microglial Genomic Atlas Zeros in on Causal AD Risk Variants (ALZFORUM, 2022)
- Can a Human Microglial Atlas Guide Brain Disorder Research? (Mount Sinai Health System, 2022)
- Polygenic Scores Paint Microglia as Culprits in Alzheimer's (ALZFORUM, 2021)

2021 **Multi-omic insights into Parkinson's Disease: From genetic associations to functional mechanisms**
Neurobiology of Disease (2021) 105580; https://doi.org/10.1016/j.nbd.2021.105580
BM Schilder, E Navarro, T Raj

2021 **Fine-Mapping of Parkinson's Disease Susceptibility Loci Identifies Putative Causal Variants**
Human Molecular Genetics (2021) ddab294; https://doi.org/10.1093/hmg/ddab294
BM Schilder, T Raj

2021 **echolocatoR: An Automated End-to-End Statistical and Functional Genomic Fine-Mapping Pipeline**
Bioinformatics (2021) btab658; https://doi.org/10.1093/bioinformatics/btab658
BM Schilder, J Humphrey, T Raj

2021 **MungeSumstats: A Bioconductor Package for the Standardisation and Quality Control of Many GWAS Summary Statistics**
Bioinformatics (2021) 37(23):4593-4596; https://doi.org/10.1093/bioinformatics/btab665
A Murphy, **BM Schilder**, NG Skene

2021

Dysregulation of mitochondrial and proteo-lysosomal genes in Parkinson's disease myeloid cells
Nature Genetics (2021) https://doi.org/10.1101/2020.07.20.212407
E Navarro, E Udine, K de Paiva Lopes, M Parks, G Riboldi, **BM Schilder**…T Raj
News
- Mount Sinai: Fighting Neurodegenerative Disorders (Mount Sinai Health System, 2019)

2021

Phenome-wide and eQTL Associations of COVID-19 Genetic Risk Loci
iScience (2021) https://doi.org/10.1016/j.isci.2021.102550
C Moon, **BM Schilder**, T Raj, K-l Huang

2021

Genome-Wide Association Study of over 40,000 Bipolar Disorder Cases Provides Novel Biological Insights
Nature Genetics (2021) 53:817-829; https://doi.org/10.1038/s41588-021-00857-4
N Mullins, AJ Forstner, KS O'Connell, B Coombes, JRI Coleman…**BM Schilder**… et al.
News
- Researchers identify 64 regions of the genome that increase risk for bipolar disorder (EurekAlert, 2021)
- Largest Bipolar Disorder Genetics Study Doubles Genetic Risk Factors (Nordic Society of Human Genetics and Precision Medicine, 2021)

2020

Tensor decomposition of stimulated monocyte and macrophage gene expression profiles identifies neurodegenerative disease-specific trans-eQTLs
PLOS Genetics (2020) 16(9):e1008549; https://doi.org/10.1371/journal.pgen.1008549
S Ramdhani, E Navarro, E Udine, AG Efthymiou, **BM Schilder**, M Parks, A Goate, T Raj

2019

Evolutionary shifts dramatically reorganized the human hippocampal complex
Journal of Comparative Neurology (2019) 528(17):3143-3170; https://doi.org/10.1002/cne.24822
BM Schilder, HM Petry, PR Hof

2019

FAIRshake: Toolkit to Evaluate the Findability, Accessibility, Interoperability, and Reusability of Research Digital Resources
Cell Systems (2019) 9; https://doi.org/10.1016/j.cels.2019.09.011
D Clarke, L Wang, A Jones, M Wojciechowicz, D Torre, K Jagodnik, S Jenkins, P McQuilton, Z Flamholz, M Silverstein, **BM Schilder**…A Ma'ayan
News
- Chosen as 'Featured Frontmatter' article in Cell Systems

2019

Geneshot: search engine for ranking genes from arbitrary text queries
Nucleic Acids Research (2019) 47(W1):W571-W577; https://doi.org/10.1093/nar/gkz393
A Lachmann, **BM Schilder**, ML Wojciechowicz, D Torre, MV Kuleshov, AB Keenan, A Ma'ayan
News
- Geneshot: Piercing the Literature to Identify and Predict Relevant Genes (University of Pittsburgh Health Sciences Library System Update, 2019)
- The Future of AI at the Hasso Plattner Institute for Digital Health at Mount Sinai (Mount Sinai Health System, 2020)

2018

eXpression2Kinases (X2K) Web: linking expression signatures to upstream cell signaling networks
Nucleic Acids Research (2018) 46(W1):W171-W179; https://doi.org/10.1093/nar/gky458
DJB Clarke, MV Kuleshov, **BM Schilder**, D Torre, ME Duffy, AB Keenan, A Lachmann, AS Feldmann, GW Gundersen, MC Silverstein, Z Wang
News
- Mount Sinai Faculty Spotlight: Ma'ayan Lab (Mount Sinai Health System, 2018)

2015

Defining elemental imitation mechanisms: A comparison of cognitive and motor-spatial imitation learning across object- and computer-based tasks
Journal of Cognition and Development (2015) 17(2):221-243; https://doi.org/10.1080/15248372.2015.1053483
F Subiaul, L Zimmerman, E Renner, **BM Schilder**, R Barr

2015 **Take the monkey and run**
Journal of Neuroscience Methods (2015) 248:28-31; http://doi.org/10.1016/j.jneumeth.2015.03.023
KA Phillips, MK Hambright, K Hewes, **BM Schilder**, CN Ross, SD Tardif
News
- Monkeys on a Treadmill? A Conversation with Dr. Kimberley Phillips (Why Social Science?)

2014 **Becoming a high-fidelity - super - imitator: what are the contributions of social and individual learning?**
Developmental Science (2014) 18(6):1025-1035; http://doi.org/10.1111/desc.12276
F Subiaul, EM Patterson, **BM Schilder**, E Renner, R Barr

2014 **Working memory constraints on imitation and emulation**
Journal of Experimental Child Psychology (2014) 128:190-200; http://doi.org/10.1016/j.jecp.2014.07.005
F Subiaul, **BM Schilder**

PREPRINTS

2026 **Genetic background shapes AI-predicted variant effects**
bioRxiv (2026) https://doi.org/10.64898/2026.04.04.715328
BM Schider, Z Liu, JJ Desmairis, D Laub, F Rahimi, P Sethi, LA Pereira, M Sun, JB Kinney, DM McCandlish, J Zhou, PK Koo

2026 **orthogene: a Bioconductor package to easily map genes within and across hundreds of species**
bioRxiv (2026) https://doi.org/10.64898/2026.01.17.700094
BM Schilder, AE Murphy, NG Skene

2025 **Cell type-specific contextualisation of the human phenome: towards the systematic treatment of all rare diseases**
medRxiv (2025) https://doi.org/10.1101/2023.02.13.23285820
BM Schilder, KB Murphy, H Dash, Y Zhang, R Gordon-Smith, J Chapman, M Otani, NG Skene

2025 **anndataR improves interoperability between R and Python in single-cell transcriptomics**
bioRxiv (2025) https://doi.org/10.1101/2025.08.18.669052
L Deconinck, L Zappia, R Cannoodt, M Morgan, scverse core, I Virshup, C Sang-Aram, D Bredikhin, **BM Schilder**, R Seurinck, Y Saeys

2024 **Harnessing generative AI to annotate the severity of all phenotypic abnormalities within the Human Phenotype Ontology**
medRxiv (2024) https://doi.org/10.1101/2024.06.10.24308475
KB Murphy, **BM Schilder**, NG Skene

2024 **Navigating the Phenomic Landscape: systematic characterisation of the latent genomic space underlying all traits and diseases**
Target to Patient (2024) http://dx.doi.org/10.13140/RG.2.2.12144.26880
BM Schilder, NG Skene

2024 **Integrative multi-omics analysis of glial signatures associated with accelerated cognitive decline in Alzheimer's disease**
bioRxiv (2024) https://doi.org/10.1101/2024.08.27.24312641
E Schneegans, N Fancy, V Chau, TKD Cheung, E Adair, M Papageorgopoulou, **BM Schilder**, PM Matthews, JS Jackson

RESEARCH EXPERIENCE

- | 2026 **Genomic AI Scientist**
Standard Model Biomedicine — New York, NY, USA
• Building multimodal world models of patient health.

- | 2024

Postdoctoral Research Scientist

Cold Spring Harbor Laboratory (Simons Center for Quantitative Biology) — Cold Spring Harbor, NY, USA

- Advancing deep learning applications in genomics and biomedicine in the laboratory of Dr. Peter Koo.
- Developing a genomic foundation model to map complex genome-phenome relationships and make highly accurate, personalized disease risk predictions.

- | 2019

Lead Data Scientist

120/80 Group — New York, NY, USA

- Offers data-driven consultation services to a wide portfolio of high-profile digital healthcare, pharmaceutical and biotech companies.
- Developed a suite of proprietary softwares to extract customised business intelligence from the published literature to generate customised and interpretable reports to clients.
- Provides clients guidance on strategic AI implementation, data analysis, publication and transparency.

2020 | 2018

Bioinformatician II

Icahn School of Medicine at Mount Sinai (Department of Neuroscience / Department of Neurology / Department of Genetics & Genomics / Ronald M. Loeb Center for Alzheimer's Disease) — New York, NY, USA

- Developed machine learning systems to integrate large-scale multi-omics datasets (e.g. whole-genome sequencing, bulk and single-cell RNA-seq, epigenomics, clinical data) to uncover the molecular mechanisms underlying neurodegenerative diseases (e.g. Alzheimer's, Parkinson's, ALS).
- Computationally identified specific disease-causal variants, pathways and cell-types for subsequent functional wet lab validation (e.g. CRISPR-cas9 editing in patient-derived cell cultures, iPSCs and cerebral organoids).

2018 | 2017

Bioinformatician II

Icahn School of Medicine at Mount Sinai (Department of Pharmacological Sciences) — New York, NY, USA

- Conducted computational systems biology research. Integrated and analyzed large-scale genomic and biomedical data (e.g. Python, R, JavaScript).
- Developed evolutionary algorithm to optimize gene network kinase regulator prediction (eXpression2Kinases).
- Developed and deployed computational tools, software, databases and web applications for basic and clinical research, resulting in 3 peer-reviewed publications.

2013 | 2011

Research Assistant

The George Washington University (Department of Anthropology) — Washington, DC, USA

- Performed dissection, histology, microscopy and quantitative stereology in post-mortem primate brain tissues.
- Trained junior and senior personnel on lab protocols.

2013 | 2011

Senior Lab Manager

The George Washington University (Department of Speech, Language & Hearing Sciences) — Washington, DC, USA

- Organized and trained dozens of undergraduates to conduct weekly cognitive development research; designed and/or directly contributed to over 15 research projects in two years.

2010

Paid Research Intern

Princeton University (Princeton Neuroscience Institute) — Princeton, NJ, USA

- Investigated the neural basis of decision-making in humans.
- Recruited participants, recorded EEG and analyzed data in MATLAB.

$ GRANTS

Total (all grants): $3,049,872
Total (as primary applicant): $311,382